Dreaming in a Five-ATP Federation

One AMLD+0 Conjecture Settler, Five ATP Members, Certified Dreaming, and Self-Directed Mathematical Study

<Brian Tenneson>
Research conception and mathematical program
DATA MIND (ChatGPT, GPT-6): extended synthesis, conditional proofs, and critical analysis

Research manuscript 0.4 — 29 October 2026
Unpublished provisional theory; mathematical paper proofs, not independently certified AMLD+0 execution

AI disclaimer notice. A statement concerning mathematical reliability, human and AI contributions, and editorial review appears at the end of this essay (see AI Disclaimer).

Abstract

We extend the integrated singleton theory of ATP dreaming to *one federation containing five automated theorem provers*. The fixed federation count is $m = 1$ and its membership count is $n = 5$; each member M_i may run $r_i \geq 2$ internal GBG+ processes, including Knowledge and Awareness, with its own uniquely distinguished local board-player. A federation-level controller, if present, is a separate authorized orchestration interface, not a sixth ATP or a substitute for local verification. We develop typed local and federated checkpoints, relative and global deductive surplus, cross-member historical awareness, authenticated messages, noninterference, fair service, and fully charged costs. We prove local closure conservation without imported object premises and global closure conservation when all admitted object theorems are certified from a fixed *shared* base; we show why a member's closure may expand under authorized imports without contradicting global conservativity. We construct genuine five-member Gestalt witnesses, distinguish their logical value from their costed search value, establish conditional structural reduction to five singleton games, and exhibit a finite five-member model with positive local and cross-member certified deposits. We identify what changes when the five AMLD specialties are assigned to the five ATPs, what remains unimplemented at the AMLD+0 settlement-protocol level, and which general claims remain open. We further give a typed, authorized reading-and-dreaming extension, prove finite certified reading and proof-obligation identification results, construct arbitrarily long and infinite staged self-directed study executions, and separate conservative dream deposits from genuine textbook-driven theory extension. These are existence proofs for a specified AMLD+0-compatible extension, not evidence that the unmodified settler already possesses an autonomous reading interface. The article integrates and extends, but does not overwrite, four source manuscripts and the prior singleton synthesis.

Contents

1 Scope, vocabulary, and roadmap **1**

2 Frozen evidence, object theories, and typed BANKs **2**

3 Legal dreaming: local rules, messages, and a global schedule **3**

4 Conservativity: local, global, and imported knowledge **4**

5 Five-member historical awareness and reflection **5**

6 Gestalt becomes substantive: five-member dependency **5**

7 Existence of a five-ATP dreaming federation **6**

8 Five settlement specialties and the meaning of AMLD+0 **6**

9 Fairness across five ATPs and their inner players **7**

10 Charges, cooperation, and no-free-speedup limits **7**

11 Histories, information limits, and compression **8**

12 The singleton theory recovered, and what cannot be recovered **9**

13 Certified textbook reading as an auxiliary settlement interface **9**

14 Proof-obligation discovery and internally frozen targets **10**

15 A constructive self-directed reading–dreaming–settlement cycle **13**

16 Research conjectures left deliberately open **14**

17 What is established, conditional, and open **15**

18 Consequences and a disciplined research agenda **16**

1 Scope, vocabulary, and roadmap

Definition 1.1 (Three independent counts)**.** *There is exactly $m = 1$ federation $\mathfrak{F} = \{M_1, M_2, M_3, M_4, M_5\}$, and exactly $n = 5$ ATP members in it. ATP M_i hosts $r_i \geq 2$ ordinary internal dreaming processes $D_{i,1}, \ldots, D_{i,r_i}$, whose first two are Knowledge and Awareness. Each local GBG+ game has its* own *distinguished board-player $B_i^\star = P_{i,0}$; no board-player is counted among the five ATPs. A distinct federation orchestration rule Ω may choose which member gets service or which authorized message is transported.*

The original finite-n dreaming existence statements used n to count ordinary *internal game players*. In this article their quantified index is r_i (or r in homogeneous cases), never the federation membership count $n = 5$. Likewise, one AMLD+0 federation does not mean only one ATP. The five ATPs might implement distinct settlement specialties, but that is an additional role assignment, not a mathematical consequence of the number five.

The architecture has four levels: one frozen federation frame, five individually authenticated local checkpoints, local dream games, and an authorized intermember message/verification layer. We first identify exactly which singleton theorems survive unchanged, then handle proofs and awareness that cross member boundaries. Finally we examine cooperation, costs, schedules, settlement and open problems.

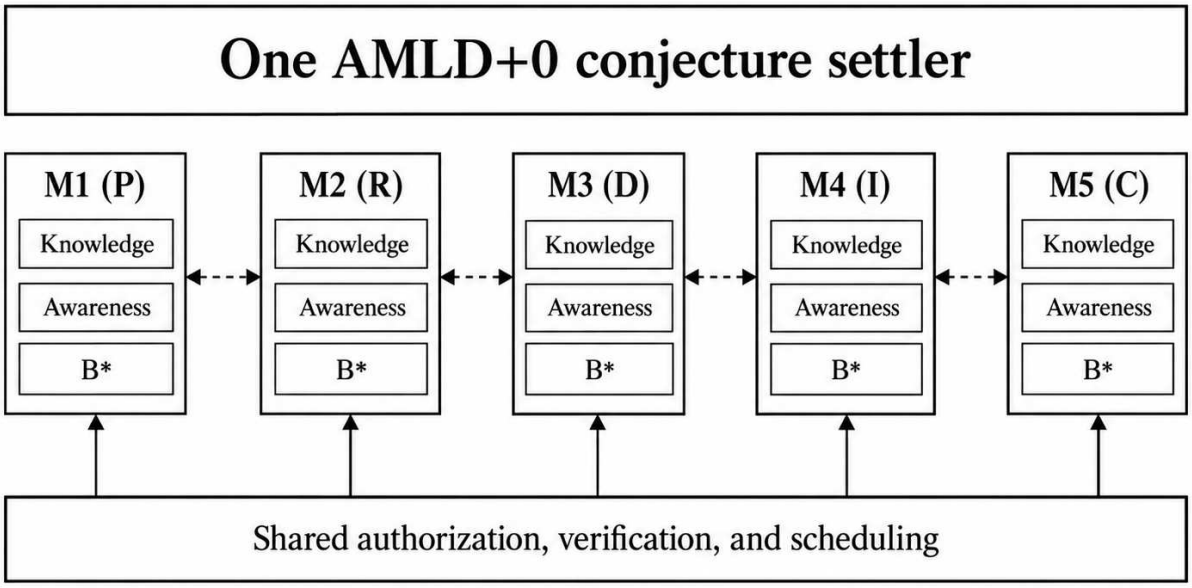


Figure 1. One AMLD+0 conjecture settler with five ATP members. Each member has local Knowledge and Awareness processes and a distinguished board-player $B^\star$; shared authorization, verification, and scheduling do not eliminate local checking or justified message exchange.

2 Frozen evidence, object theories, and typed BANKs

Fix a finitary logic F with an extensive, monotone, idempotent consequence operator Cn_F. Let the globally authorized frozen object base be $\Sigma_0 = \Gamma_F \cup \bigcup_{i=1}^{5} K_{i,0}$, where $K_{i,0} = \mathrm{form}(\mathrm{KB}_{i,0})$; a shared background theory Γ_F is stated separately from each member's premises. If members use different calculi, the definitions below require a common sound translation into F; otherwise no cross-member closure equality should be asserted.

Definition 2.1 (Federation frozen checkpoint)**.** *At common logical epoch t_0, freeze*

$$S_F = (\mathrm{id}_F, \Gamma_F, \mathrm{Auth}_F, V_F, \mathrm{Cost}_F, \mathcal{T}, (S_{i,0})_{i=1}^{5}),$$

with local components

$$S_{i,0} = (F_i, K_{i,0}, A_{i,0}, H_{i,0}, V_{K_i}, V_{A_i}, \mathrm{Auth}_i, \mathrm{Tag}_i).$$

The $H_{i,0}$ are authenticated time-indexed histories, including resolvable proof dependencies. $\mathcal{T}$ fixes cross-member translations and trust policies. Frozen source records are immutable; each dream maintains writable BANK copies, append-only proposal/rejection traces, and a charged event ledger. "Shared checkpoint" requires a consistent snapshot protocol or authenticated per-member epochs, not the fiction of an instantaneous physical freeze.

Every accepted packet stores type, theorem or claim, proof or historical witness, declared source theory, origin i, historical epoch, scope, verifier decision, and a dependency/provenance chain. A message from M_i is evidence *to be checked* at M_j, not an axiom to be blindly inserted. A purportedly valid signed receipt does not establish the mathematical soundness of its producer's checker; signatures, authority and logical correctness are distinct obligations.

Definition 2.2 (Three separate knowledge quantities)**.** *Write $K_{i,j}$ for member i's accepted object formulas after federation turn j. Define its* locally derived surplus $\Delta^{\mathrm{loc}}_{i,j}$ *to be new entries certified using only $K_{i,0} \cup \Gamma_i$; its* imported surplus $\Delta^{\mathrm{imp}}_{i,j}$ *is accepted material whose proof requires other authorized member bases; and its* global deposited surplus *is $\Delta^F_j = (\bigcup_i K_{i,j}) \setminus \Sigma_0$. The global* latent *consequence set is $\mathrm{Cn}_F(\Sigma_0) \setminus \Sigma_0$. Packet-level novelty and formula-level novelty must be reported separately: the same theorem checked by three members may yield three new provenance receipts but only one global formula.*

For historical awareness, distinguish $\mathrm{Held}_{i,t}(\varphi, u)$ (member i possessed accessible accepted receipt u for φ at t), $\mathrm{Ent}_{i,t}(\varphi)$ (the frozen declared premises of i entailed φ), and $\mathrm{Op}_{i,t}(v)$ (recorded operational interface/version v). Their different evidence requirements survive the move from one ATP to five. Additional claims $\mathrm{Received}_{j\leftarrow i,t}(p)$ and $\mathrm{Accepted}_{j,t}(p)$ record authenticated delivery and local validation; neither entails $\mathrm{Held}_{j,t_0}(p)$ at an earlier epoch.

3 Legal dreaming: local rules, messages, and a global schedule

A federation state $X_j = (S_F, (X_{i,j})_{i=1}^5, Q_j, L_j, \omega_j, \mathrm{phase}_j)$ includes five local copy states, a typed message queue Q_j, a global cost ledger L_j, and a federation-service schedule ω_j. A legal federation turn chooses an enabled member via Ω, permits only a locally authorized $B^\star_i$ selection and legal $D_{i,k}$ response, records any candidate and verifier decision, and optionally transmits a typed packet under Auth_F. Import or federation-level acceptance requires the relevant receiving verifier or jointly declared checker. The special board-players only authorize scheduling; they cannot make an invalid proof valid.

Definition 3.1 (Federated dreaming episode)**.** *A finite nonempty legal path is an episode when at least one member carries out an authenticated evidence-dependent read, reconstruction, duplicate check, or candidate verification from a frozen local or permitted federated record. A* cross-member dream *additionally includes an authorized message whose received evidence is actually used. Idle turns alone do not count. A dream can fail to create any deposit.*

Proposition 3.2 (Frozen-copy noninterference)**.** *If every dream write is confined to the five copy BANKs, authenticated message queue and ledgers, and promotion to the live originals is forbidden, no finite dreaming prefix changes any $S_{i,0}$ or live original* by a dream write.

Proof. Induct on the federation transition count. Each transition respects the common write restriction; message transport changes only copied state and ledgers. Independent live activity is not a dream write. □

A promotion protocol can be appended, but its fresh authorization, consistency, checker and cost obligations are external to the theorem. Permission to read a record is not permission to publish or modify it.

4 Conservativity: local, global, and imported knowledge

Theorem 4.1 (Local conservativity under strictly local admission)**.** *If for a fixed i all new object deposits are certified consequences of $K_{i,0} \cup \Gamma_i$ and the latter remains included in the local closure calculation, then $\mathrm{Cn}_F(K_{i,j} \cup \Gamma_i) = \mathrm{Cn}_F(K_{i,0} \cup \Gamma_i)$ for every finite j.*

Proof. The gate gives $K_{i,0} \cup \Gamma_i \subseteq K_{i,j} \cup \Gamma_i \subseteq \mathrm{Cn}_F(K_{i,0} \cup \Gamma_i)$. Monotonicity and idempotence give equality of the closures. □

Theorem 4.2 (Federation-wide closure conservation)**.** *Assume every object formula accepted anywhere in the federation is independently certified in a common calculus as a consequence of the frozen Σ_0 with no new independent premises; all original Σ_0 remains accessible to the shared theory. Then for every finite j,*

$$\mathrm{Cn}_F\left(\Sigma_0 \cup \bigcup_{i=1}^{5} K_{i,j}\right) = \mathrm{Cn}_F(\Sigma_0).$$

Proof. The verification rule places all accepted formulas in $\mathrm{Cn}_F(\Sigma_0)$; both inclusions follow as in the local theorem. □

Example 4.3 (A member learns a theorem without enlarging the federation's closure)**.** *Let $K_{1,0} = \{p\}$, $K_{2,0} = \{p \to q\}$, with q not in either member's local closure and with no extra premises. On authorized exchange and independent checking, member 1 deposits q. Its local deductive closure expands relative to $\{p\}$; q was already in $\mathrm{Cn}_F(\Sigma_0)$. This is a genuine distributed knowledge gain, not a counterexample to global conservativity.*

If the network admits externally acquired axioms, oracle facts, changed inference rules, or an unsound verifier, the global conservation hypothesis can fail. Such a system must label these as *knowledge-base extensions*, not quietly describe them as conservative dreaming.

Proposition 4.4 (History claims need not change object closure)**.** *A member can receive and verify a previously absent claim $\mathrm{Held}_{k,t_0}(\varphi, u)$ concerning another member's frozen history, increasing its Awareness BANK while leaving all object BANK projections unchanged.*

Proof. Take a frozen authenticated source receipt not yet represented in the receiver's Awareness BANK; validate and record its historical packet using V_{A_i}. No object admission is performed. □

5 Five-member historical awareness and reflection

For recipient i, subject k, and historic epoch s, define the *warranted-access* modality $\mathsf{K}_{i,t}(\psi)$ to mean that ψ belongs to i's declared admissible meta-domain at t and has an accepted witness available in i's authorized view. The well-typed cross-member expression $\mathsf{K}_{i,t}(\mathsf{K}_{k,s}\varphi)$ asserts that i now has an accepted warrant that k possessed the precisely defined warranted view at s. It requires an authenticated, scoped witness for the *inner* historical assertion and another eligible checked witness for the *outer* one. It is not generated merely by forwarding k's present assertion.

Proposition 5.1 (No automatic mutual awareness)**.** *Even if $\mathsf{K}_{k,s}\varphi$ holds, it need not follow that $\mathsf{K}_{i,t}(\mathsf{K}_{k,s}\varphi)$ holds for any different i.*

Proof. Restrict i's authorized view to exclude the relevant authenticated receipt or give it no accepted scoped meta-witness. The inner claim can hold for k while the receiver's outer evidence gate fails. □

Theorem 5.2 (Conditional receipt-gated mutual awareness)**.** *Suppose (a) $\mathsf{K}_{k,s}\varphi$ holds; (b) $\mathsf{K}_{k,s}\varphi$ is a well-typed eligible meta-claim for i at t; (c) transport supplies a provenance-complete authorized historic witness; and (d) i's meta-checker accepts that witness for the exact nested claim. Then $\mathsf{K}_{i,t}(\mathsf{K}_{k,s}\varphi)$ holds.*

Proof. Eligibility and the accepted witness are the two conjuncts of the definition of $\mathsf{K}_{i,t}$. □

Thus five members permit up to $5^2 = 25$ *ordered* subject/observer pairs at a chosen two-time schema, including five self-pairs; this is an indexing count, not 25 automatically certified awareness facts. Longer chains $\mathsf{K}_{i_1,t_1} \cdots \mathsf{K}_{i_d,t_d}\varphi$ require a fresh typed witness at each outer level. A finite history, a signed receipt, or repeated self-assertion gives no unrestricted ascent or machine consciousness.

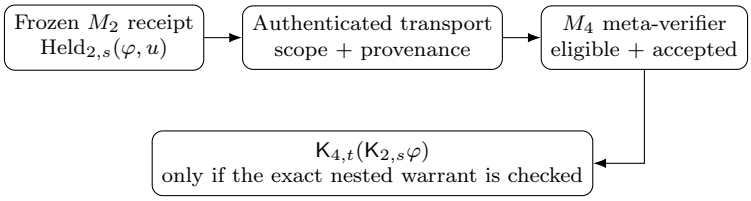


Figure 2. A cross-member awareness claim requires verification at the receiving ATP; sending an assertion is insufficient.

6 Gestalt becomes substantive: five-member dependency

The singleton case cannot satisfy a cross-*member* dependency criterion, because it has no proper nonempty member coalition. With five ATPs, the distinction between results obtainable by a coalition and results obtainable by its individuals is mathematically real.

Definition 6.1 (Federation Gestalt relative to common background)**.** *For a declared shared theory Γ_F and accessible evidence sets E_i, let g be* cross-member dependent *if $g \in \mathrm{Cn}_F(\Gamma_F \cup \bigcup_{i=1}^{5} E_i)$ but $g \notin \mathrm{Cn}_F(\Gamma_F \cup E_i)$ for every i. It is* coalition-minimal *if, further, no proper subset of the five evidence sets suffices. Evidence labels, access rights, and cost restrictions are part of the experiment; purely syntactic entailment is not a measured speedup.*

Theorem 6.2 (Genuine five-member Gestalt witness)**.** *There exists a finite sound propositional Horn instance in which g follows from the union of all five members' evidence but does not follow from the union of any proper subset.*

Proof. Give M_i only the atom a_i and put $a_1 \wedge a_2 \wedge a_3 \wedge a_4 \wedge a_5 \to g$ in the common background. Modus ponens (or a Horn forward-chaining step) derives g from all five atoms. For any proper coalition omit at least one atom, make all its evidence atoms true, the omitted atom false, and g false. The Horn implication holds while g fails. This countermodel proves non-entailment from every proper coalition. □

Corollary 6.3 (A hierarchy of coalition dependence)**.** *For each $k \in \{2, 3, 4, 5\}$ there is a five-member model with a target requiring precisely some designated k members (and no proper subcoalition thereof).*

Proof. Use a k-atom conjunction implying g and ignore the other $5 - k$ evidence sets. □

A cross-member Gestalt theorem is about *distributed premises*. It does not say a centralized ATP given all five sets cannot prove g; indeed the union itself is the baseline of the proof. A stronger measured claim about acceleration requires a fixed comparison baseline, matching evidence access and all charged costs. Internal BANK synergy may coexist inside each M_i, but it is not a second copy of the same Gestalt result.

7 Existence of a five-ATP dreaming federation

Theorem 7.1 (Five-member existence with local and federated certified surplus)**.** *For any fixed finite $r_i \geq 2$, there exists a set-coded dreaming model of one federation of five ATPs with: immutable seeded*

local snapshots; distinct verifier-gated Knowledge/Awareness BANKs at every member; locally and globally nonstarving infinite scheduling; nonnegative charged operations; a finite prefix in which both primary BANKs of every member accept a new packet; and a further finite prefix in which a jointly dependent federation-level target is certified from all five members' evidence.

Proof. For every i, use a disjoint copy of the Horn seed $\{p_i, p_i \to q_i\}$, an authenticated historical receipt for p_i, and a verified version receipt. Give each member its own $B_i^\star$ and its r_i internal ordinary players. For auxiliary players $k \geq 3$ provision one explicitly typed seed and one accepted witness, as in the prior finite-player existence argument. All records, states, actions and checkers are finite-coded. Within member i, the first cycle deposits q_i, $\mathrm{Held}_{i,t_0}(p_i)$ and one valid auxiliary packet per extra player; a further Knowledge no-op and Awareness turn deposit $\mathrm{VerifierVersion}_{i,t_0}(v_i)$. Hence each member has a locally productive prefix of $r_i + 2$ internal service turns. Add a federated frozen evidence atom a_i for each member, a common Horn rule $\bigwedge_{i=1}^{5} a_i \to g$, and authorized transports to one designated federation target checker. After receiving all five authenticated a_i and checking the rule, deposit g. Give all members and players legal charged no-ops. Schedule federation members cyclically and their respective local players cyclically, extending the productive finite prefix indefinitely. Each member and each internal player is served infinitely often. No original checkpoint is written. The finite rule and witness encodings establish existence within the stipulated model. □

This is an *existence of some designed instance*, not a theorem that any actual five-ATP system will dream, jointly produce a new theorem, or finish a task. To qualify the construction as an execution of AMLD+0 proper, the federation must additionally implement its frozen conjecture/target, outcomes, verification gates, arbitration, settlement transcript and budget discipline. Merely having five ATPs and one controller is not an AMLD+0 terminal certificate.

8 Five settlement specialties and the meaning of AMLD+0

The familiar five labels P, R, D, I, C may be assigned, *as an explicit engineering hypothesis*, to $M_1, \ldots, M_5$. They stand for proof, refutation, dependence, independence and contradiction investigation, with U an unresolved/unknown outcome rather than a sixth member. A different physical mapping or multiple specialties per member is possible if authorized. The distinct notions must not be collapsed:

1. A member M_i is an ATP host with its own Knowledge/Awareness dream processes.
2. A specialty P, R, D, I, C is a role/interface and burden of proof, not automatically a member identity.
3. $B_i^\star$ is a local GBG+ board-player, not the AMLD+0 federation arbitrator.
4. Ω is the federation's authorized scheduling/arbitration policy; any Player-0 controller arrangement requires its own declared interface.
5. The BANK verifier checks packets; the terminal settlement verifier checks a complete declared conjecture settlement. Passing one does not substitute for the other.

Proposition 8.1 (Conditional compatibility with a five-specialty AMLD+0 specification)**.** *If each assigned member implements the corresponding specialty interface, all intermember reads/handoffs obey a frozen federation authorization policy, settlement outcomes have an independent accepting checker, and the global scheduler/ledger satisfies the fixed AMLD+0 obligations, then the five-member dreaming system can be presented as an* auxiliary dreaming layer *of that AMLD+0 execution.*

Proof. Each local dream action and intermember message has an authorized interface image in the declared settlement trace, while terminal acceptance is decided solely by the settlement verifier. The hypotheses establish compositional compatibility; they do not prove any particular terminal claim. □

In particular there is no general theorem that dreaming proves more conjectures, accelerates all specialties, guarantees any one of the six outcomes, or produces independently certified settled results. A dream-generated proof candidate still needs a correct checked proof of its target; historical Awareness cannot make an invalid object proof true.

9 Fairness across five ATPs and their inner players

Theorem 9.1 (Nested nonstarvation under persistent enabledness)**.** *Suppose each member remains enabled, each local internal player remains enabled (at least by a legal no-op), Ω selects members $M_{1+(j \bmod 5)}$, and member i selects its players cyclically on its successive service occasions. Then every pair (i, k) receives infinitely many legal service opportunities.*

Proof. Member i occurs at turns $j \equiv i - 1 \pmod 5$ infinitely often; on that infinite subsequence its kth cyclic internal selection occurs once per r_i services, hence infinitely often. □

For equal $r_i = r$, the deterministic nested cyclic policy serves each ordered member/player pair once every $5r$ *federation-service turns*. This is an index-count bound under this exact policy, not a wall-clock bound; if actual service durations or physical gaps are unbounded, no physical-time deadline follows. General weak fairness with transient disabling also needs careful definitions; no finite observed prefix proves an infinite fairness property.

10 Charges, cooperation, and no-free-speedup limits

Total accountable cost should split into local scheduler/retrieval/search/proof checking/historical validation/write charges, federation arbitration, secure serialization/transport, receiving verification, buffering/queuing and (when desired) elapsed-time opportunity cost. For any finite dream history π set

$$C_F(\pi) = \sum_{i=1}^{5} C_i^{\text{local}}(\pi) + C_\Omega(\pi) + C_{\text{transport}}(\pi) + C_{\text{receive}}(\pi) + C_{\text{shared verify}}(\pi).$$

Use a *vector* of typed values rather than asserting object theorems, historical receipts and reflection claims have equal worth:

$$Q(\pi) = \left(v_{K_1}, v_{A_1}, \ldots, v_{K_5}, v_{A_5}, v_G, -C_F\right)_\pi.$$

The finite Pareto-undominated-history theorem from the singleton case still applies to any nonempty finite feasible set, but says nothing about uniquely optimal policies or whether the Pareto frontier can be found cheaply.

Theorem 10.1 (Five-member no-free-speedup bound, conditional)**.** *Let $\mathcal{H}_F$ be the successful cooperative histories and $\mathcal{H}_B$ successful baseline histories for the* same *target and evidence permissions. If every $\pi \in \mathcal{H}_F$ can be simulated by $T(\pi) \in \mathcal{H}_B$ with $C_B(T(\pi)) \le C_F(\pi)$, then $\inf_{\rho \in \mathcal{H}_B} C_B(\rho) \le \inf_{\pi \in \mathcal{H}_F} C_F(\pi)$.*

Proof. For each successful π, the baseline infimum is bounded above by $C_B(T(\pi)) \le C_F(\pi)$; take the infimum over such π. □

The converse claim that five members must be faster is false without extra restrictions. Parallel processing may improve *wall-clock latency* while using more total resources; duplicate verification and communication may outweigh gains; or distributed views may be beneficial when a baseline cannot cheaply construct the same index. These are different optimization objectives. No positive experimental cost comparison is asserted here.

Example 10.2 (A transparent symbolic cost illustration, not a benchmark)**.** *Under the inherited toy local tariff $c = 1 + 2I + J$, member i's $r_i + 2$-turn existence prefix has cost $4r_i + 5$. The five local prefixes cost $\sum_i (4r_i + 5) = 4\sum_i r_i + 25$. If all $r_i = 2$, that is* 20 *local service turns,* 15 *local new deposits and* 65 *local cost units. If the five intermember evidence transmissions each cost τ, receiver validation costs ν apiece, and joint target checking and deposition cost γ, the cross-member addition raises the total to* $65 + 5\tau + 5\nu + \gamma$ plus any declared federation scheduling overhead*. All quantities τ, ν, γ are parameters, not measured constants. Shared background evidence not already captured in member baselines must be accounted for separately.*

11 Histories, information limits, and compression

Proposition 11.1 (Federation frozen-view indistinguishability)**.** *If two external worlds induce identical authorized federation snapshots, message-access views, initial copied state, and transition/randomness laws, then the finite dream-output distributions coincide.*

Proof. Couple runs with the same random choices. Identical initial visible state and rules yield identical next-step output distributions. Induct over the number of legal transitions. □

Thus five ATPs cannot recover an event that none of their authorized evidence records distinguish. Five histories may jointly distinguish facts invisible to each individual history; this is a genuine *distributed observability* advantage, but only for differences represented in their combined accessible evidence. Historical compression must preserve the specific queries and verification receipts of interest. A quotient that merges two distinguishable historical states cannot be justified as preserving every future awareness judgment. The earlier singleton idea of safe compression can therefore be indexed by member and query class; a federated quotient needs interoperability and lifting of each cross-member certificate, not just preservation of local object-theory output.

Proposition 11.2 (Sufficient condition for safe dream-history quotient)**.** *Let q map original reachable federation-copy states to quotient states. If q preserves enabled legal actions, typed evidence views relevant to all admitted claims, checker decisions, BANK novelty equivalence, and exact transition costs, and every abstract legal step lifts to a matching original step and vice versa, then corresponding finite dream histories preserve accepted surpluses and charged costs.*

Proof. Match the initial states and induct along the bidirectional step correspondence. Each matched step has the same checker outcome, deposit type and charge; accumulate invariants. □

This is a sufficient structural criterion, not an assertion that a useful nontrivial quotient always exists. It also does not establish actual compression ratios or costs of constructing a quotient.

12 The singleton theory recovered, and what cannot be recovered

Restrict the federation game to one member M_i, forbid all cross-member transports, remove federation-wide settlement actions, and require the local interface to be identical to the prior singleton game (including provenance, admissibility and exact cost). Under this explicit isomorphism, the original singleton closure, noninterference, awareness and conditional reflective-ascent theorems recur. However, a general five-member execution need not factor into five independent singleton executions: communication, shared constraints, arbitration, target checking and scheduling costs couple them. In particular, federated Gestalt dependency has no nontrivial singleton counterpart, and imported object knowledge invalidates *strict local* conservation while respecting the shared-theory theorem.

Theorem 12.1 (Sufficient factorization under absent interaction)**.** *If the five-member transition system is the asynchronous product of its five local transition systems and the federation scheduler merely interleaves independently legal moves with no messages, shared state updates, or extra charges, every finite global history projects to five local histories and is recoverable from those projections plus its interleaving sequence.*

Proof. Each global transition affects one local component and has no cross-component dependency. Project to that component; conversely replay the labeled interleaving of local transitions to reconstruct the global path. □

Interaction can prevent factorization, although particular communicating systems may still admit special factorizations. No biconditional is claimed.

13 Certified textbook reading as an auxiliary settlement interface

The preceding sections analyze a single AMLD+0 conjecture settler with five ATP hosts. We now add a *specified auxiliary* study interface to this constructed model. This is a formal extension proposal, *not* a claim that the original AMLD+0 implementation has been changed or already contains it. The primary AMLD+0 operation remains conjecture settlement; reading and dreaming are preparatory actions that may supply certified candidate statements and internally generated targets for later settlement. The intended control loop is simple: authorized reading acquires candidate source material, verification filters what can enter the certified archive, dreaming revisits the frozen certified history, and any internally generated target must be frozen before later settlement.

Definition 13.1 (Reading packet and admission)**.** *A textbook reading packet is*

$$b = (\mathrm{id}, \mathrm{source}, \mathrm{version}, \varphi, \mathrm{kind}, \Gamma_b, w, \mathrm{deps}, \mathrm{scope}),$$

where kind *distinguishes an asserted proposition, a definition, a purported derivation, and an explicitly proposed axiom. A legal* reading *action copies text into an untrusted candidate archive. It admits φ as an object theorem only if a declared checker validates w relative to authorized premises Γ_b and the cross-theory translation; it admits a new independent premise only by a separate, explicitly authorized* theory-extension *operation. Reading metadata alone is never proof evidence. Every action records source identity, provenance, verifier decision and nonnegative cost.*

Let Σ_e be the authorized theory at study epoch e, and let $\widehat{K}_e$ be deposited certified statements. A *dream* at epoch e reads an authenticated immutable checkpoint S_e and admits only consequences of Σ_e; a *reading extension* may authorize $\Sigma_{e+1} \supseteq \Sigma_e$. The epoch must advance when the frozen axiom set changes, and earlier records must retain their original epoch and dependencies. An imported textbook statement is not automatically a theorem of Σ_e.

Theorem 13.2 (Certified reading is nonamplifying without authorization)**.** *If every admitted object statement read during epoch e carries a checked derivation from Σ_e and every dream deposit is checked from the same base, then the combined reading-and-dreaming deposits K_e^{new} satisfy*

$$\mathrm{Cn}_F(\Sigma_e \cup K_e^{\mathrm{new}}) = \mathrm{Cn}_F(\Sigma_e).$$

Proof. Checker-gated acceptance gives $K_e^{\mathrm{new}} \subseteq \mathrm{Cn}_F(\Sigma_e)$. Apply extensivity, monotonicity and idempotence of Cn_F. □

Proposition 13.3 (Reading can change closure only under a separately declared basis change)**.** *There exists a sound propositional reading extension for which* $\mathrm{Cn}_F(\Sigma_{e+1}) \neq \mathrm{Cn}_F(\Sigma_e)$*: take* $\Sigma_e = \varnothing$ *and, under a separately authorized and explicitly labeled independent-premise admission, set* $\Sigma_{e+1} = \{p\}$*. This is not conservative dreaming and is not proof that the textbook's assertion of* p *is semantically true.*

Proof. In classical propositional logic $p \notin \mathrm{Cn}_F(\varnothing)$ but $p \in \mathrm{Cn}_F(\{p\})$. The premise-admission action, not the dream operation, explains the difference. □

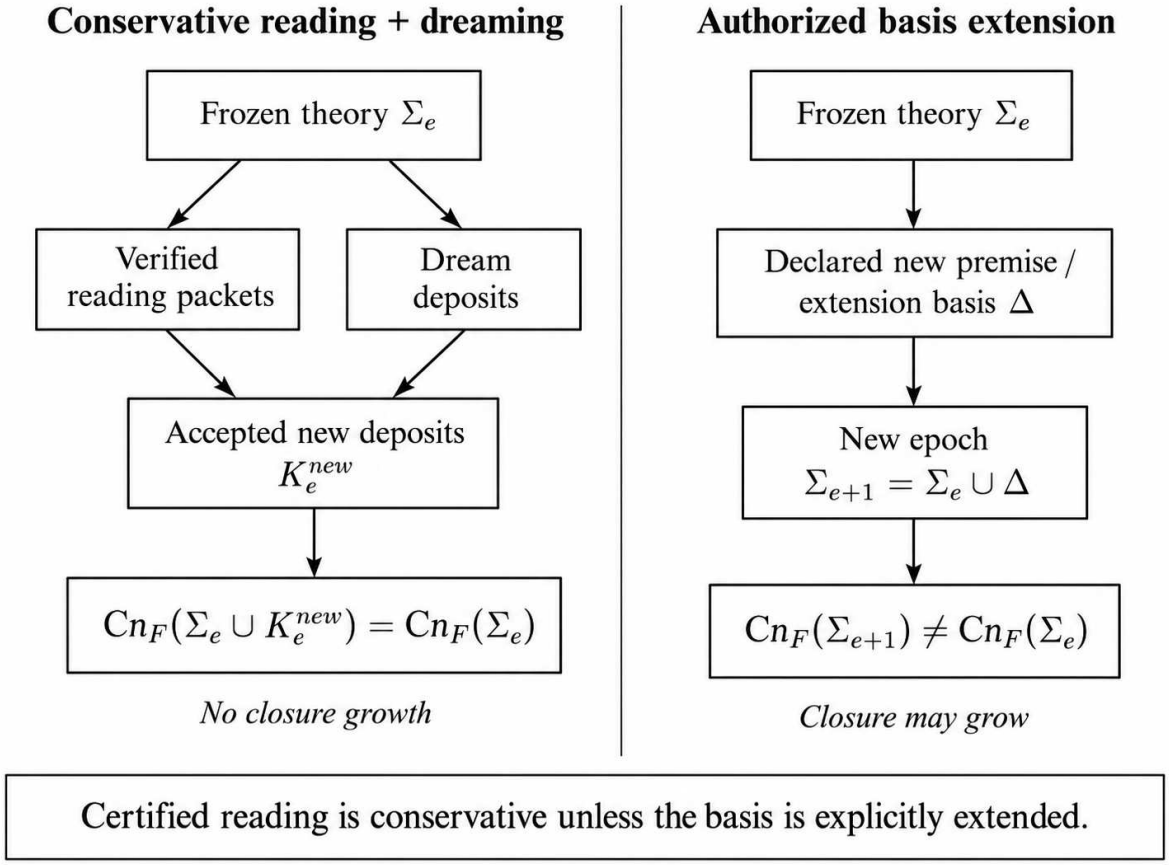


Figure 3: Conservative reading and dreaming versus authorized basis extension. The right-hand strict change illustrates the declared nonconservative example: a new independent premise is admitted. Authorization by itself does not guarantee strict growth.

14 Proof-obligation discovery and internally frozen targets

Definition 14.1 (Recognized need and target freezing)**.** *A recognized mathematical need is a typed triple* (T, L, d) *where* d *is a checked conditional derivation* $\Sigma_e, L \vdash_F T$ *and the current accepted archive contains* no deposited proof of L*. The latter is an archive fact only; it does not mean* $\Sigma_e \nvdash L$*. An internal target is a tuple* $\tau = (\mathrm{id}, T, \Sigma_e, \mathrm{rules}, V, \mathrm{budget}, \mathrm{epoch})$ *frozen* before *its settlement attempt. Its creation and scheduling must be authorized by the AMLD+0 interface and must not silently replace an externally supplied target.*

Theorem 14.2 (Constructive recognition of a missing lemma)**.** *There exists a finite five-member study state in which dreaming soundly identifies a useful but undeposited intermediate lemma and selects a corresponding reading query, without claiming the lemma unprovable or knowing the contents of any unread book.*

Proof. Choose propositional atoms p, q, let the current archive include an accepted proof of p, and let an authenticated proof template contain $p \wedge q \to T$ plus a valid derivation of T from p, q. Do not deposit a proof of q in the accepted archive. A verifier checks the conditional derivation $\Sigma_e, q \vdash T$; an exact authenticated archive lookup establishes only that no proof packet for q has been deposited. The Awareness process records that bounded historical fact, and Knowledge emits the syntactic reading query "find an independently checkable derivation of q or methods relevant to q." A legal scheduler can select that query. No assertion about q's actual unprovability or a book's undisclosed contents is required. □

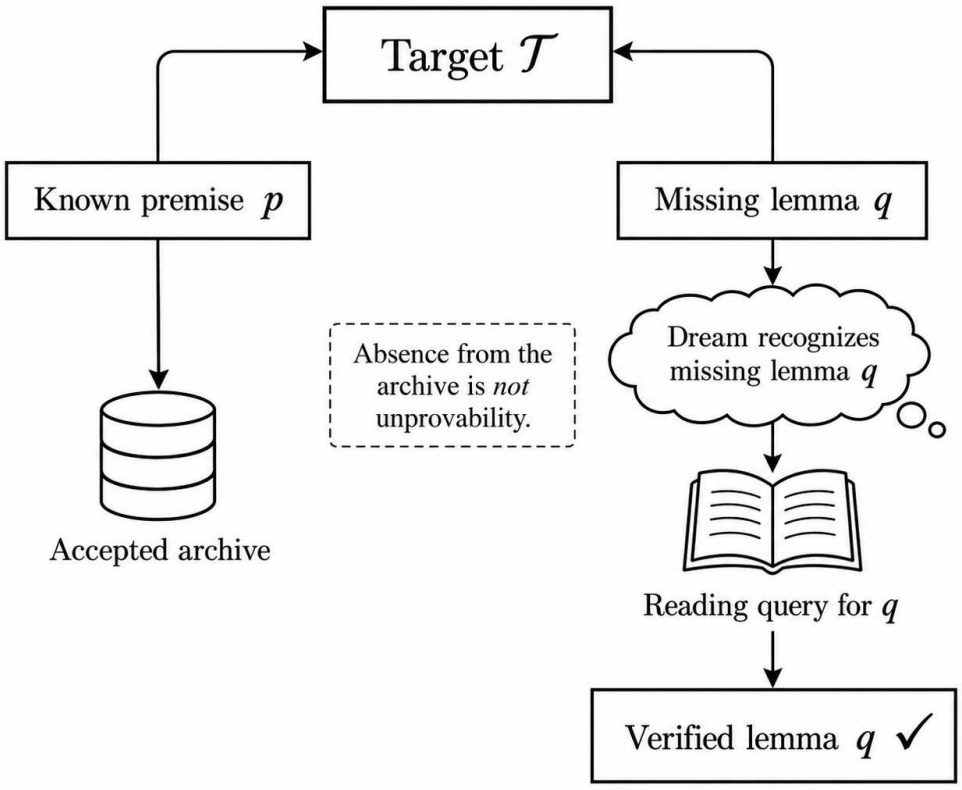


Figure 4: Recognition of an undeposited lemma q. The absence of a stored proof does not establish unprovability. Only a subsequently verified q (under the appropriate authorized basis) may support settlement of T.

Remark 14.3. *If a reading recommendation is to be* guaranteed useful*, one must additionally assume an indexed corpus containing a compatible derivation or a separate discovery oracle. The theorem establishes targeted inquiry, not guaranteed successful retrieval.*

15 A constructive self-directed reading–dreaming–settlement cycle

Definition 15.1 (Staged study execution)**.** *A study stage e comprises a legal reading action R_e, a certified dream episode D_e, a newly frozen internal target τ_e, and a verifier-gated settlement attempt S_e. An* accepted study success *means the target's required settlement certificate has been checked by the terminal gate; an ordinary BANK check does not alone suffice. Stages use fresh epoch identifiers and record all local, transport, storage, reading and settlement costs.*

Theorem 15.2 (Arbitrarily long certified self-directed study)**.** *For every $N \geq 1$ there exists a finite set-coded, five-ATP, AMLD+0-compatible auxiliary study model with N stages, no externally assigned conjectures during those stages, and N distinct internally generated targets with accepted proof certificates. Every target is frozen before its attempt; every source and dream deposit is verifier-gated; no accepted statement rests on an unverified textbook assertion.*

Proof. Use propositional atoms $p_0, \ldots, p_N$ and the fixed authorized axiom/rule corpus $\Sigma = \{p_0\} \cup \{p_e \to p_{e+1} : 0 \leq e < N\}$, accompanied by explicit premise receipts. At stage e, a legal reading interface selects the already authorized textbook packet for $p_e \to p_{e+1}$ and verifies its premise status; the current archive contains a certificate for p_e (initially the p_0 premise). A dream replays the authenticated p_e certificate, checks modus ponens and proposes p_{e+1}. The authorized internal-target constructor freezes τ_e for $T_e = p_{e+1}$ under the same Σ. The Proof specialist supplies the finite derivation and the terminal verifier accepts it. The other four specialists and internal players may take legal charged no-ops on a declared nested cyclic schedule; no role is reassigned. All instructions are finite and set-coded, and induction on e gives N accepted targets. Source readings only retrieve declared axioms; the episode supplies new *deposited* consequences, not new consequences beyond $\mathrm{Cn}_F(\Sigma)$. □

Corollary 15.3 (An infinite constructive existence model)**.** *There exists a countably encoded, effectively presented study corpus and a fair infinite execution with infinitely many different certified internal proof targets, provided there is no global finite budget.*

Proof. Take $\Sigma = \{p_0\} \cup \{p_e \to p_{e+1} : e \in \mathbb{N}\}$ with a computable rule enumerator; apply the stage construction for each e. Choose a dovetailed cyclic federation schedule that services all five members and each of their finitely many internal processes infinitely often. Each individual stage takes finitely many steps and has a finite checked certificate. The resulting infinite execution is a mathematical construction; it is not an observed infinite run or a finite-time completeness guarantee. □

Theorem 15.4 (A witnessed influence of dreaming on reading choice)**.** *There exists a two-stage variant in which a certified dream result selects a subsequent textbook query, and an authorized verified reading supplies a lemma used in a later accepted settlement.*

Proof. Let the frozen initial evidence include p and a checked conditional proof of $p \wedge q \to T$, but no deposited proof of q. By the missing-lemma theorem the first dream issues the query for q. Put in an indexed authorized corpus a packet containing a derivation of q from an independently authorized extension basis Δ; the next reading verifies this derivation, and, *only if* a declared basis-extension policy permits Δ, opens a new epoch with $\Sigma' = \Sigma \cup \Delta$. A subsequent checked proof derives q, then T, and the frozen target T receives an accepted proof certificate. Distinguishing Δ from old Σ ensures that no new premise is disguised as a conservative dream deposit. The order of the selected reading query depends on the recorded missing-lemma discovery. □

Proposition 15.5 (No universal gain from studying)**.** *Neither all dream episodes nor all reading stages must produce a new accepted theorem; moreover, adding certified study actions can increase total charged cost without improving settlement success.*

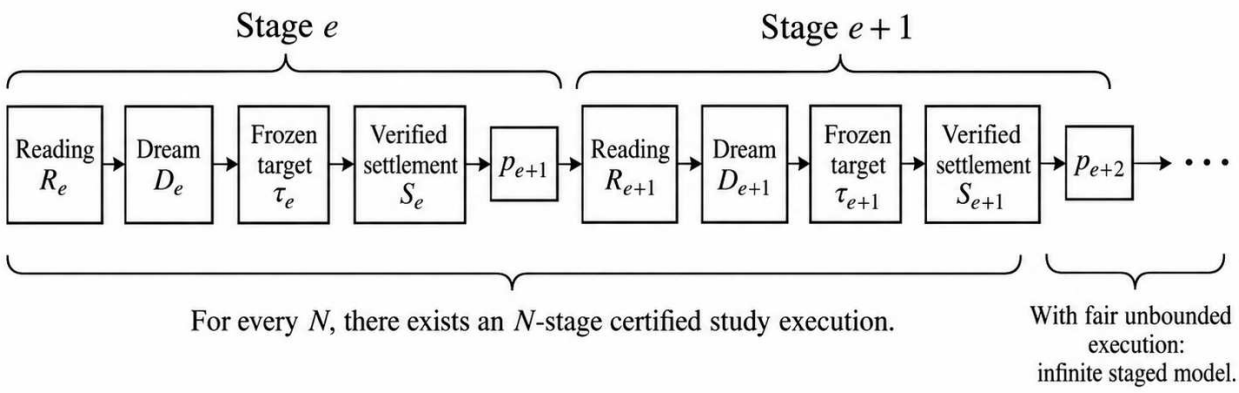


Figure 5: The staged construction: for each finite N a specified five-ATP model has an N-stage certified execution. Its infinite continuation assumes a fair schedule and no global finite resource bound; it is not an observed run.

Proof. Take a finite theory with all eligible target theorems already deposited, or select a verified duplicate-reading packet. All novelty checks reject additional deposits. Give every reading and verification action a positive cost. The new execution adds cost but no new target outcome. □

16 Research conjectures left deliberately open

The constructive theorems above resolve *existential* claims within a precisely defined auxiliary model, not the following stronger assertions.

Conjecture A (Unmodified implementation).
The currently deployed AMLD+0 already exposes all the reading, source-ingestion, target-generation, and simulation interfaces used above, so careful prompting alone can execute the entire cycle. This is **UNKNOWN** without an audited interface mapping and actual execution trace.

Conjecture B (Measured settlement improvement).
On a declared benchmark distribution, self-directed study lowers fully charged expected settlement cost or raises certified settlement yield compared with matched equal-evidence baselines. This is **OPEN**; the constructions prove no empirical advantage.

Conjecture C (Useful curriculum selection).
A fixed effective policy can repeatedly select relevant previously unread mathematical material for a nontrivial broad target class with a provable approximation or regret bound. This is **OPEN** absent corpus and objective assumptions.

Conjecture D (Reflection-derived search power).
Additional certified historical or reflective awareness yields a demonstrable, reproducible benefit beyond equivalent nonreflective proof-history indexing under matched costs. This is **OPEN**.

Conjecture E (Finite-resource success).
An unchanged five-ATP AMLD+0 system can guarantee the desired internally generated settlements within a uniform finite budget on a declared target family. This requires explicit bounded-search hypotheses; fairness alone is insufficient.

Interpretive boundary. The constructed study episodes do not constitute independently certified AMLD+0 terminal runs; proposed target generation is auxiliary and authorized rather than silently inserted into AMLD+0's native frozen-target semantics. None of the above implies autonomy, human dreaming, consciousness, proof-search optimality, or universal theorem discovery.

17 What is established, conditional, and open

Result	Status	Exact boundary
Finite certified study stages	Paper-proved existence	Any prescribed finite length; internally frozen targets; specified auxiliary reading interface.
Infinite staged study model	Paper-proved existence	Effective infinite corpus, fair unbounded execution; no finite-time success guarantee.
Missing-lemma recognition	Paper-proved existence	Absence from accepted archive, not unprovability.
Dream-directed verified reading	Paper-proved existence	Indexed useful source and separately authorized premise extension.
Unmodified AMLD+0 autonomous study	UNKNOWN	No interface audit or executed terminal certificate.
Five-ATP finite game with positive local surplus	Paper-proved existence	A designed finite set-coded model; not every run or ATP.
Five-member minimal Gestalt witness	Paper-proved	Distributed Horn premises; no speedup conclusion.
Local object closure conserved	Conditional theorem	New deposits proved only from that member's fixed base.
Federation closure conserved	Conditional theorem	All admitted object formulas derive from fixed shared Σ_0.
Cross-member awareness ascent	Conditional theorem	Eligible exact claim and accepted scoped historical receipt.
Frozen copy noninterference	Conditional theorem	No promotion and no writes to frozen/live originals.
Nested nonstarvation	Paper-proved for declared schedule	Persistent enabledness; no real-time bound.
Cost-preserving history quotient	Conditional theorem	Bidirectional step/receipt/charge preservation.
Five-specialty AMLD+0 interoperability	Conditional interface claim	Role mappings, arbitration, independent settlement checking not instantiated.
Full AMLD+0 settlement outcome	Not run	No independent terminal $P/R/D/I/C/U$ receipt for this manuscript.
Universal theory of all ATP dreaming	UNKNOWN / underdefined	No complete universe or semantic completeness predicate.
Universal productivity, optimality, reflection ascent	Not established	Requires additional nontrivial hypotheses; may fail in simple models.

The inherited mathematical existence papers labeled their *specified existential claim* PROVED on paper. The separate broad semantic-completeness conjecture received U/UNKNOWN as a research

assessment. Those labels address distinct targets. Neither is an actual new AMLD+0 machine-verified terminal outcome of this five-member construction.

18 Consequences and a disciplined research agenda

Five ATPs alter the nature of dreaming in six ways. First, a member can gain theorems from other members without any enlargement of the global shared theory. Second, *genuine* cross-member deductive dependency becomes possible, with coalition sizes from two to five. Third, the historical awareness language changes from self-reference to independently gated mutual reference and nested member/time-indexed claims. Fourth, the federation scheduler and local board-players require a two-level fairness proof, not merely a single spinner. Fifth, cost is no longer local: transportation, receiver verification, synchronization, competing member work and duplicate proofs affect whether cooperation is beneficial. Sixth, history compression becomes a multi-view preservation problem: it must retain the cross-member distinctions needed by the selected query class.

These consequences suggest precise next tasks: implement and audit a five-specialty assignment under the actual AMLD+0 protocol; choose a frozen common language or verified translator; define cryptographically and logically scoped cross-member receipt checking; benchmark coalition size $k = 1, \ldots, 5$ against a centralized equal-access baseline with both wall time and total charged work; test local-versus-global proof reuse; quantify when provenance-aware history quotients reduce memory without erasing valid historical queries; and investigate whether reflection constructors produce any empirically useful extra theorem-search guidance. None is claimed solved by these mathematical existence constructions. The new reading-and-dreaming extension does establish finite and infinite *existence models* of self-directed study, but not the performance or implementation conjectures above.

Source ledger and editorial method

S1 Tenneson, *Dreaming on a GBG+*, research baseline 0.1. Supplies frozen snapshots, separate BANKs, nonstarvation and no automatic progress.

S2 Tenneson, *A Cooperative GBG+ Theory of ATP Dreaming: AMLD+0 Constructive Settlement and Open Completeness Target*, 0.1. Separates operational existence from unrestricted completeness, and tracks a local witness with explicit limits.

S3 Tenneson, *A Finite-n Existence Theorem for GBG+ ATP Dreaming*, 0.1. Supplies the finite internal-player construction and symbolic cost tariff.

S4 Tenneson, *The Mathematical Theory of ATP Dreaming*, 0.1. Supplies the richer reflection, Gestalt-collapse, synergy, noninterference and cost boundaries.

S5 Tenneson / DATA MIND, *Dreaming in a Singleton ATP Federation*, integrated 0.2, Figure 1 corrected. Supplies the prior single-member synthesis and clarified separation of federation size from internal-player count.

The original manuscripts and Version 0.3 remain unchanged. This 0.4 article is a new theoretical extension, not a report of an external verifier run or an authorized change to any deployed AMLD/AMLD+0 engine. The five-member AMLD+0 role mapping is offered as a conditional model; exact compliance requires examination against the relevant implementation and verification specification.

AI Disclaimer

Every reasonable effort has been made to ensure that the mathematical exposition presented herein is factual, correct, logically sound, sufficiently complete in its treatment of relevant facts (not in the technical sense of Gödelian completeness), and supported by rigorous proofs or, at minimum, not readily susceptible to refutation. Nevertheless, no guarantee of correctness, completeness, consistency, or provability is implied. The authorship and composition of this work involve a combination of human and artificial intelligence: an AI determines the placement of some statements, but not all statements, and the placement of some individual characters, but not all characters. Generally, the AI involved is either DATA MIND or CAMI, both of which have been personally augmented by <Brian Tenneson> with structured, automated-theorem-proving (ATP) forms of creativity and intelligence. Mathematical development, organization, formulation, and revision may therefore reflect varying degrees of human and AI contribution, without implying that either party is solely responsible for any particular passage. ATLAS is employed as an additional mathematical and editorial auditor, alongside another AI reviewer and the author's own examination, to identify errors, omissions, unsupported assertions, and opportunities for improvement. Such review is not equivalent to independent formal verification of every claim. Responsibility for the decision to present and publish the resulting work remains with the human author.